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PROJECT PROPOSAL

Title: Predictive Market Demand Analysis in E-Commerce Using Machine Learning

1. Introduction

The E-Commerce industry has grown rapidly, generating large amounts of data related to customer behavior, pricing, and sales. Analyzing this data is important for understanding market demand and making better business decisions.

Traditional methods are often not enough to handle such complex data. Therefore, Machine Learning techniques are used to identify patterns and predict future demand more accurately.

This project focuses on analyzing e-commerce market demand using data-driven approaches to support effective decision-making.

2. Problem Statement

In the E-Commerce industry, accurately understanding and predicting market demand is challenging due to large and complex datasets involving pricing, sales, and customer behavior. Traditional analysis methods often fail to capture hidden patterns and trends.

Without proper demand analysis, businesses may face issues like overstocking, understocking, and inefficient pricing strategies. Therefore, there is a need to use Machine Learning techniques to analyze data and predict demand more effectively.

3. Objectives

- To analyze market demand patterns in the E-Commerce sector using historical data.
- To identify key factors influencing demand, such as pricing and sales trends.
- To apply Machine Learning techniques for demand prediction.
- To develop a model that improves decision-making for pricing and inventory management.

4. Model Used

This project uses two models to predict market demand in the E-Commerce sector.

- **Linear Regression:** A simple model used to identify the relationship between input features and demand. It acts as a baseline.
- **Support Vector Regression (SVR):** An advanced Machine Learning model that captures non-linear patterns and improves prediction accuracy.

Both models are compared to determine the better-performing approach.

5. Proposed System

The proposed system uses a data-driven approach to analyze and predict market demand in the E-Commerce sector.

It collects historical data such as pricing and sales, performs preprocessing, and applies Machine Learning models like Linear Regression and SVR to predict demand.

The system then evaluates model performance and provides insights to support better decision-making in pricing and inventory management.

6. Tools and Technologies

- **Programming Language:** Python
- **Platform:** Google Colab
- **Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn
- **Techniques:** Machine Learning, Data Analysis, Data Visualization
- **Dataset:** E-commerce pricing and revenue dataset

7. Expected Outcomes

- Accurate prediction of market demand in the E-Commerce sector.
- Identification of key factors affecting demand, such as pricing and sales trends.
- Improved decision-making for pricing and inventory management.
- Performance comparison between models to select the best approach.

8. Conclusion

This project uses Machine Learning to predict market demand in the E-Commerce sector. The models provide insights that help improve pricing and decision-making.